PROBLEM 3

Sitting Sales Representatives

Actual Sales Call District				
Rep's Base	1	2	3	4
1	1	4	5	7
2	4	1	3	5
3	5	3	1	2
4	7	5	2	1

CONSTRAINTS- B19:E19 = BINARY

	Hours Available	160
	Fixed cost	88000
	Variable Cost	80000

Monthly Call Requirements

	District 1	District 2	District 3	District 4	Total
Calls Required	50	80	100	60	290

Number of Rep assigned to each district

	District 1	District 2	District 3	District 4	Total
Number of Rep	0	0	5	0	=SUM(B17:E17)

DECISION VARIABLES- B19:E19, B23:E26, B17:E17

CONSTRAINTS- B17:E17<= B20:E20 , B17:E17 = INTEGER,

F23 : F26 <= G23: G26

Base	District 1	District 2	District 3	District 4	Hours used	Hours Available
District 1 Rep	0	0	0	0	=B23*B6+C23*C6+D23*D6+E23*E6	=H6*B17
District 2 Rep	0	0	0	0	=B24*B7+C24*C7+D24*D7+E24*E7	=H6*C17
District 3 Rep	50	80	100	60.0000000000001	=B25*B8+C25*C8+D25*D8+E25*E8	=H6*D17
District 4 Rep	0	0	0	0	=B26*B9+C26*C9+D26*D9+E26*E9	=H6*E17
Total calls taken	=SUM(B23:B26)	=SUM(C23:C26)	=SUM(D23:D26)	=SUM(E23:E26)		
Calls Required	50	80	100	60		

CONSTRAINTS - B27:E27 = B28:E28

Calculation of Annual Cost

	District 1	District 2	District 3	District 4	Total
Fixed Cost	=B19*H7	=C19*H7	=D19*H7	=E19*H7	=SUM(B32:E32)
Variable Cost	=B17*H8	=C17*H8	=D17*H8	=E17*H8	=SUM(B33:E33)

TOTAL COST	**=SUM(F32:F33)**	OBJECTIVE- MINIMISE